

Judge Q&A — Every Question, Answered Simply

SaResQ · SIH26177 · built against the problem statement, line by line · read this on the way in

Where every file is

WHAT	PATH
Deck script — 12 slides + 3 reserve, with what to say	~/saresq/docs/SaResQ_Deck_Draft.pdf
This Q&A sheet	~/saresq/docs/SaResQ_QA_Prep.pdf
Full technical report (19 pp, glossary)	~/saresq/docs/SaResQ_Model_Architecture_Report.pdf
Research paper — 14 pp, 26 equations, 12 tables, 10 figures	~/saresq/docs/paper/sareshq_paper.pdf
3D interactive model (real CAD, explodable)	claude.ai artifact — docs/assembly_3d.html
Build guide	claude.ai artifact — docs/assembly_guide.html
Live ground-station demo	http://sareshq.local:8091 (payload) / :5050 (dashboard)
On-Pi benchmark data	~/sareshq/results/pi_benchmark.json

If you remember one paragraph, remember this

"Drone thermal search already works — what fails is the human watching the feed. We put the watching onboard: a 3.2 MB model on a ₹4,000 Raspberry Pi that finds 85% of people in aerial thermal imagery, in real time, using 71% of its frame budget. It bolts to any 450 mm quadcopter, never touches the flight controller, and tells the ground exactly how sure it is and why."

0 · The problem statement, bullet by bullet — say this FIRST if asked "what did you actually build"

SIH26177 lists seven "Expected Solution" capabilities. A technical judge has this list in front of them. Do not wait for them to find the gaps — name each one, then say which bucket it's in. **BUILT** = working and measured. **PARTIAL** = a real, honest version of it exists but not the full ask. **SCOPED OUT** = deliberately not built, and here is why that was the right call for a one-person, six-week, ₹14,000 build.

Autonomous Navigation — GPS + GPS-denied, SLAM, obstacle avoidance

SCOPED OUT — BY DESIGN

We do not fly the drone. We treat the airframe and its flight controller (ArduPilot/PX4, a commodity product) as a solved problem, and we build the *payload that rides on it* — the part nobody else on this problem statement is likely to have measured numbers for. Flying, SLAM and obstacle avoidance are a different, extremely mature field; reinventing them with one person and six weeks would have meant a worse flight controller and a worse payload. We plug into any drone a department already owns and flies.

On-Device AI Inference — real-time, no cloud

BUILT

Everything runs on the Raspberry Pi itself. No internet call, ever, in the detection path. 28.2 ms per crop, worst case 177 ms of a 250 ms budget, measured on the board.

Multi-Sensor Fusion — RGB + thermal + IMU + GPS

BUILT

All four sensors feed one verdict per candidate. Thermal finds body heat, motion confirms it's alive, the RGB camera confirms it's a person-shape when there's light to see by, and GPS+IMU turn the pixel into a map coordinate. See Section H for exactly how this combines.

Hazard Classification — floods, fires, smoke, debris, unstable structures, landslides, chemical leaks

PARTIAL — NAME THE FIVE CLASSES

The onboard hazard classifier is trained on **AIDER** and reports five scene classes: fire, flood, collapsed building, traffic incident, normal — **95.2%** accurate. We do not have a class for exposed electrical lines, chemical leaks, or landslide specifically — no public labelled dataset for those exists at the scale we needed, and inventing labels ourselves would be a guess dressed as a measurement. **Say this straight: "five real hazard classes, measured at 95%, and three more from the list we chose not to fake."**

Geo-Tagged Mapping — survivor + hazard locations, safe-access routes

PARTIAL — HONEST, NOT COMPLETE

Every detection with a GPS fix is placed on a live satellite/street map, with the payload's own position and a following camera view. What's missing on purpose: we never draw a bearing to a target, because the payload carries no compass — only range is a real measurement, so a contact draws as a *ring* at its measured distance, not a fake pin. Safe-access-route planning is not built.

Emergency Alerting — automatic alerts, prioritised rescue recommendations

PARTIAL

The verdict ladder (Section H) IS the prioritisation — every candidate is ranked from **LIVE_PERSON** down to **CLEAR**, and the ground station's review queue sorts by that automatically. What's missing: push notification / SMS to a phone. That is a small integration, not a research problem, and is next on the list.

Offline Resilience

BUILT

Detection, fusion and the verdict ladder run with zero network. The ground map has a baked offline tile set for the demo area. The link between payload and ground station degrades to "last seen 8 s ago" rather than losing data when radio drops — nothing is lost, nothing is invented.

The one-line summary if a judge asks "so what's actually in scope here"

"We didn't try to build a drone that flies itself AND sees for itself in six weeks — we picked the half of this problem statement that was under-served, built it properly, and measured every number we're about to show you."

A · The problem, and why it matters

Q Why does this need to exist? Drones with thermal cameras already do this.

They do — and every one of them streams video to a human who has to watch it. That person gets tired. A survivor is a 2.6-pixel warm smudge on a 32×24 thermal image at 4 frames a second. A tired human can't rewind, can't combine three weak sightings from three passes, and can't put a GPS coordinate on something they half-saw. We automate the *looking*, not the flying.

Q Who actually uses this? Who pays for it?

State disaster response forces and municipal fire services — organisations that already own drones and already fly them at incidents. The pitch is not "buy a drone"; it's "make the drone you own able to search on its own for ₹14,000."

Q What's the real-world impact, in one sentence?

It turns "someone glanced at the video" into a number: how sure we are, why, and where — which is what lets a coordinator send a team instead of hoping.

B · "Isn't this just...?" — differentiation

Q Isn't this just YOLO running on thermal images?

If it were, it wouldn't run. A whole-frame detector needs about 320 ms per frame on a Pi 4 — over the 250 ms budget before it does anything else. The contribution is the **cascade**: a cheap statistical check looks at every frame for free and points at 2–6 small patches; the network only ever looks at those.

Measured: 177 ms worst case, budget 250 ms. That's the whole system in one number.

SAY IT SIMPLY Airport security. Everyone walks through the cheap metal detector; only the few who beep get the expensive search. Our thermal gate is the metal detector.

Q Why not just buy a DJI thermal drone?

A DJI M30T is roughly ₹10–15 lakh and still **streams to a person**. It's a better camera on the same bottleneck. We're a ₹14,000 module that adds autonomous perception to drones a department already owns.

Q What is genuinely new here — in plain words?

- **The cascade makes it fit on cheap hardware.** Look cheap everywhere, look hard only where something's warm.
- **Evidence adds up across passes.** Three weak sightings of the same spot combine into one confident detection. A live video feed can't do that — it has no memory.
- **The gate works both directions.** Cold anomalies count too — a body in water, a void under rubble. In afternoon sun, hot roof tiles can be warmer than a person; a "find hot spots" system fails exactly then, ours doesn't.

C · Compute, frame rate, and long flights

Q How does the compute keep up with the frame rate, for the whole flight — not just a demo minute?

Because almost nothing runs on most frames. The thermal sensor reads at 4 Hz. Every one of those frames gets the cheap check — that's free, it's just arithmetic on 768 numbers. The expensive part, the neural network, only switches on for the 2–6 small patches the cheap check flagged. Most frames in a quiet scene fire the network on zero or one patch. There's no "running out of steam" over a long flight, because the workload per second doesn't grow with flight time — it only grows with how many warm things are in view *right now*.

Q What if the whole frame is full of warm things — worst case?

We measured that, not guessed it: **6 crops plus the hazard classifier, back to back, is 177 ms.** The budget per frame is 250 ms. So even the worst case we could construct still leaves 29% headroom. That's on the actual Raspberry Pi 4, not a laptop.

Q Does it slow down or overheat on a long flight?

On the bench: 42.8 °C, no thermal throttling, 95 MB of RAM used out of 905 available. Nothing in the design gets slower with elapsed time — there's no growing log kept in memory during flight, no cache that fills up. It runs the same speed at minute 1 and minute 40. **What we have not yet measured is a real 20–30 minute flight** — only bench soak tests. That's an honest gap, not a hidden one.

Q What about battery — does the AI compute drain the drone's flight battery?

No — it can't, by construction. The whole payload runs off its own separate power bank, not the drone's flight battery. It has no electrical connection to the airframe at all. Longer flights cost the drone's own battery the same as if the payload weren't thinking at all; our payload just has to last as long as its own power bank, independently.

D · How fusion actually works — simply

Q Explain fusion like I've never heard the word before.

Four separate signals get checked for the same spot on the ground, and they either agree or they don't:

- **Thermal** — is something here warmer or colder than its surroundings, the way a living body would be?
- **Motion** — did that warm patch move between frames? Moving proves alive. Not moving proves nothing — an unconscious person doesn't move either, so we never punish stillness.
- **Vision** — when there's enough light to see by, does it actually look like a person?
- **Position** — GPS + the drone's tilt (IMU) turn "this pixel" into "this GPS coordinate", so three sightings from three different passes can be recognised as the same spot.

These four don't get smooshed into one blurry average. They climb a ladder — the more of them agree, the higher the verdict:

VERDICT	WHAT IT MEANS
LIVE_PERSON	thermal + motion + vision all agree — highest confidence
LIVE_BODY	warm AND moving — alive, whatever the camera can or can't see
PERSON	looks like a person, and thermal backs it up
BODY_HEAT	body-temperature and still — may be unconscious, or the camera just can't see (night, smoke)
VISUAL_ONLY	looks like a person but NO thermal support — could be a poster, a mannequin, a coat on a chair; verify before acting
HEAT	a warm spot that isn't behaving like a body
CLEAR	nothing found

SAY IT SIMPLY A stock camera-only detector will happily box a photograph of a person, or a mannequin. Body heat plus motion is two separate physical facts agreeing that something is actually alive — that's worth more than one camera's opinion, so it outranks it.

Rule we do not break: a MISSING measurement (camera can't see, no motion yet) is never reported as a NEGATIVE one ("no person"). At night the camera is normally blind — that's not a failure, it's the reason the payload is thermal-first. The console says the reason out loud, e.g. "likely life — thermal only, camera blind (too dark)".

Q Why not a neural network to do the fusion, instead of a ladder of rules?

Two reasons, both practical. **It has to be explainable** — a rescue coordinator needs to be told *why* something was flagged, in words, not "the network said 0.83". **It has to be trustworthy under repetition** — evidence gets added up across multiple flyover passes, so the confidence number has to behave like an honest probability, not drift. A hand-tuned ladder with a documented reason for every rank gives us both; a black box gives us neither.

E · How it integrates with the drone — simply

Q How does this actually attach to a drone? Walk me through it.

It's a self-contained box that bolts on — **340 g**, four M3 bolts into a standard vibration-damping plate, or straps through built-in slots. Think of it like clipping an action camera to a drone: it rides along, it does not plug into the flight computer at all.

- Its own battery (a power bank) — not the drone's flight battery.
- Its own GPS and its own motion sensor (IMU) — not the flight controller's.
- Its own onboard computer (Raspberry Pi) doing all the thinking.
- Zero wires to the autopilot, zero wires to the flight battery.

That last point is the whole safety argument in one sentence: this payload cannot crash the drone, because it has no electrical path to anything that flies it. Fitting it is a mechanical job — four bolts — not an avionics integration a certified technician has to sign off on.

Q Does it need any modification to the drone or its firmware?

No. It reads the world with its own sensors and never talks to the flight controller. Any 450 mm-class quadcopter (F450/S500 size) or bigger that can lift 340 g extra already works. That was a deliberate design constraint from day one — "no room for error" meant no touching a system that keeps the aircraft in the air.

Q How does the ground team get the results while it's flying?

A radio link (in the demo, Wi-Fi) carries live stats at 1 Hz and confirmed events every 3 s to a ground-station dashboard — a map, a live feed, an evidence log. If the link drops, nothing is lost: the payload keeps recording locally and the ground station just shows "last seen 8 s ago" until it reconnects. **We also built one more hop after that:** the ground station can re-run a second, larger detector on anything the payload flagged, using the laptop's own compute — a bigger model gets a second look at borderline cases without needing to fit on the drone at all.

F · Defending the numbers

Q 85% — is that good?

For a 3.2 MB model on a Raspberry Pi, yes. It's not state-of-the-art, and we don't claim it is. A model fifty times larger on a GPU would beat it. The constraint IS the contribution: this has to run on hardware a relief agency can buy fifty of.

SAY IT PLAINLY "Person AP50 of 0.850 — it finds and correctly boxes about 85 of every 100 people in the imagery it was built for."

Q Did you try the newer YOLO11 model? Why not use it?

We did — trained it end to end on our own data, same recipe, same epochs, and it did not win. Person AP50: 0.8249 for YOLO11n vs 0.8211 for our shipped v8n — a difference of 0.004, smaller than the run-to-run noise we measured (± 0.029). We wrote the adoption rule *before* running the test, so we couldn't talk ourselves into switching after seeing a result we liked. It's 14% fewer parameters but our accuracy is bounded by how many pixels a person occupies on the thermal sensor, not by model capacity — so a smaller model with the same capability doesn't help here, and we kept the one already deployed and tested.

Q What's your AP50-95? (the sharp question)

0.371 — it finds people reliably and boxes them loosely. For this mission that's the right trade: we need "is there a person, and where" — not a traced outline. At 14.7 mm per pixel, a box 5 px loose is 7 cm on the ground. No rescue outcome depends on 7 cm.

Q Is that measured on the model you actually deploy?

Yes, and it matches the reference. The 3.3 MB file on the Pi scores 0.860; the full-precision reference scores 0.850. We test that comparison directly, because it once caught a real bug.

Q What was the bug?

Standard 8-bit quantisation was silently running our model at 32% of its real accuracy — 0.2315 instead of 0.8229. It loaded fine, ran fast, drew sensible-looking boxes. Nothing threw an error. Cause: full INT8 also quantises internal activations, including a box-decoding constant, which came out slightly wrong — a 1–2 px shift that destroys a 50%-overlap match on a 9–45 px person. Fix: keep weights at 8-bit, activations at float (called w8a32). Same 3.3 MB file, 99.2% of reference accuracy.

SAY IT SIMPLY Like saving a photo as a heavily compressed JPEG — looks fine on screen, measure it and a third of the detail is gone. We only knew because we measured.

Q Any other bugs you found and fixed?

Two, both found while re-measuring numbers for the research paper — good sign we keep checking our own work:

- **Torn thermal frames.** The thermal sensor reads out in two interleaved halves; roughly **half of all frames** can be read mid-refresh, so the two halves come from slightly different moments. That inflates noise and makes a "confident detection" threshold rise to compensate. Rejecting torn frames raised our measured detection significance from about 3.5σ to $4\text{--}6\sigma$ on the same people.
- **A motion detector that invented movement.** Comparing two genuinely still frames, our alignment step returned tiny spurious shifts (noise, not real motion) and then treated that noise as movement — 170 out of 600 still frames (28%) were flagged as "moving" when nothing moved. That matters because a moving body is promoted higher on the verdict ladder. Fixed with a small deadband; false alarms dropped to 1 of 600, and it still catches a real 4-pixel limb movement.

Both fixes are shipped, tested, and documented in the paper — not swept under the rug.

G · Architecture choices

Q Why a thermal camera with only 32×24 pixels? That's nothing.

Because it never has to identify anybody. At 20 m a prone adult is about 2 thermal pixels — there's no texture to learn, only contrast. So the thermal stage is a "something's here" test, not a classifier. Identification is the visible camera's job, on the crop the thermal already pointed at.

SAY IT SIMPLY A smoke detector doesn't identify what's burning — it just says "something here, go look." That's our thermal stage.

Q Why P3–P5 and not P2? Small objects need high resolution.

We built both and measured. P2 lost every object-size bin at twice the compute.

BIN	P3	P2
8–16 px	0.657	0.612
16–32 px	0.772	0.753
32+ px	0.814	0.791
GFLOPs	8.86	17.43

The deployed path runs on a 160 px crop anyway, where a prone adult is 115×31 px — that's not a small-object problem at all by the time the network sees it.

Q You used pretrained weights. Isn't that cheating?

It's standard practice, and with ~7,000 training images it's the only sensible choice. The detector starts from COCO weights, then we trained the whole network — not just the last layer — on thermal aerial data.

SAY IT SIMPLY Someone who can already read learns a new language faster than someone learning to read from scratch. We didn't teach it what an edge is; we taught it what a human looks like in thermal, from above.

H · The ground station, the map, and honesty rules

Q What does the ground station actually show?

A live map (satellite or street view) centred on the payload, a live thermal + RGB feed, an evidence log per detection, a hazard page, and a review queue ranked by confidence. Detections appear as icons near the drone as they're confirmed; the map follows the drone automatically unless someone pans away from it manually.

Q How do you avoid showing a rescuer a location that's wrong or made up?

A short list of rules we do not weaken, even for a nicer-looking demo:

- **No fix, no pin.** If GPS hasn't locked, nothing is drawn as a precise point. An operator's manual guess or the browser's own Wi-Fi location draws dashed and clearly labelled — never mistaken for the drone's real position.
- **No fake bearing.** The payload has no compass, so we only know *how far* a warm target is, not which direction. We draw that as a ring at the measured range — never a pin at a guessed angle, because that would look more precise than it is.
- **Empty means empty.** No demo data, no simulated targets, unless a `--simulate` flag is on, and even then those tracks are visibly marked as simulated.

I · Cost, scale, regulation

Q Break down the cost.

Roughly ₹10,000–14,000: Raspberry Pi 4 ~₹4,000 · MLX90640 thermal ~₹4,000 · Pi Camera ~₹1,200 · power bank ~₹1,000 · IMU + GPS ~₹800 · printed parts ~₹300 of filament · cabling and fasteners. No licence fees, no cloud costs — it runs entirely offline.

Q **Can this scale to manufacturing?**

Every part is commodity and in stock. The only custom item is the printed tray, which is parametric CAD — a design change is a one-line edit and a 12-second rebuild, and at volume it would be injection-moulded.

Q **DGCA regulations?**

It's a passive payload with no flight authority, so it doesn't change the aircraft's certification basis — the operator flies under whatever rules already apply to their drone. **Be honest: we have not sought a formal opinion, and we'd do that before deployment.**

J · The hard questions — answer these honestly

Rule for this whole section

Say the weakness first, in plain words, then what you'd do about it. A judge who finds a flaw you hid stops trusting everything else you said. A judge you hand a flaw to starts trusting you more.

Q Have you actually flown this?

No. It's benchmarked on the target hardware, on the bench — not in the air. Vibration, real flight attitude, and real thermal scenes over rubble are all unvalidated. That's the single biggest gap and we know it.

Q What doesn't work yet?

- **Fusion in the researched/tracked sense is weak** (ROC AUC 0.670 on the richer 18-feature model) — several of its inputs need the sensors mounted and flying before they can even be measured. The live payload uses the simpler verdict ladder instead, which is deliberately more conservative.
- **No daytime visible-only detector.** Detection is thermal-first today; the visible camera confirms, it doesn't lead.
- **No autonomous navigation, SLAM, or obstacle avoidance** — see Section o. Scoped out on purpose.
- **Three of seven hazard types from the problem statement aren't classified** — chemical leaks, exposed electrical lines, landslide-specific — no dataset exists for us to measure them honestly.
- **Fully buried victims aren't detectable** — thermal and RGB are both line-of-sight. LIDAR / ground-penetrating radar for that is our identified next hardware step, not built.
- **Never tested on a real deceased person** — see Section J's dead-body question. We reason from physics, not from a measurement we don't have.

Q Can it see someone buried under rubble, or completely under a collapsed wall?

No — and no thermal or visible camera can. Both are line-of-sight: they only see the surface facing the sky. If a person is completely under a wall or under rubble with no gap facing up, there is no heat signal and no image for us to receive at all — this isn't a software limit, it's what the sensors physically can, and cannot, see through.

We find people who are exposed or partly exposed — on the surface, in a gap, at the edge of a debris pile. **For a fully buried person, you need a sensor that doesn't rely on line-of-sight** — ground-penetrating radar or LIDAR-based structural mapping, or acoustic/seismic listening for tapping and breathing. That's real hardware, not a software update, which is why we've scoped it as **our next step after this build**, not something we're claiming today.

Q Will it detect dead bodies? How?

Honestly: we have not tested this on an actual deceased person, so we won't claim a number we don't have. Here's the physics, in simple terms, so the reasoning is clear even without that test:

- **A body that's still warm** (recently deceased) would still trigger our thermal gate — heat is heat, the sensor can't tell warm-and-alive from warm-and-recently-not, by temperature alone. It just wouldn't show *motion*, so our system would rank it the same as an unconscious living person: a body-heat signature with no movement, flagged for a human to verify in person. We do not and cannot claim to tell the difference — that call is for the responder, not the AI.
- **A body that has cooled to match its surroundings** (enough time has passed) would look like the background to a thermal sensor — no contrast, no detection. This is a limit of thermal imaging generally, not something specific to us.

The honest line to give a judge: "We haven't tested on a real deceased person and won't guess an answer we haven't measured. What we can say from the physics: a recently deceased, still-warm body would likely be flagged the same way an unconscious living person is — warm, not moving, verify in person — and a body that's cooled to ambient temperature would not be detected by thermal at all. We treat that as an open question for real-world validation, not something to claim either way."

Q How much of this did you actually build yourselves?

Be specific and honest. Datasets and pretrained backbones are public. What's ours: the cascade architecture, the thermal gate, the verdict ladder, the deployment runtime, the quantisation study, the parametric CAD, the ground station, the live map and its honesty rules, and the validation harness — **223 automated tests**, plus two real bugs we found in our own system and fixed while re-checking our numbers.

K · Numbers to have on the tip of your tongue

THING	NUMBER	CONTEXT
Person AP50, HIT-UAV	0.850	the headline
Person AP50, second dataset	0.544	proves generalisation
AP50-95	0.371	loose boxes, fine for rescue
Hazard classifier	95.2%	5 AIDER classes
YOLO11n vs v8n, matched recipe	+0.004	below noise — kept v8n
Model size	3.2 MB	w8a32
Inference, 160 px crop	28.2 ms	on the Pi
Worst-case frame	177 ms	of a 250 ms budget
RAM used	95 MB	of 905 available
Torn thermal frames	~50%	found + rejected, sigma rose 3.5→4-6
Motion false alarms, before/after fix	170/600 → 1/600	real bug, found and fixed
Payload mass	340 g	450 mm-class quad
Cost	₹14,000	bill of materials
Best altitude	25 m	26 m swath
Tests passing	223	

L · Analogies that do a lot of work

FOR	SAY
The cascade	Airport security — everyone through the cheap metal detector, only the few who beep get searched.
The thermal gate	A smoke detector. It doesn't identify what's burning; it says "something here, go look."
The quantisation bug	Saving a photo as a heavy JPEG. Looks fine on screen; measure it and a third of the detail is gone.
Torn thermal frames	A video call where the picture updates half at a time — you can still tell someone's there, but measuring exactly how warm they are needs the whole frame from one instant.
Transfer learning	Someone who can already read learning a new language, versus learning to read from scratch.
Payload/drone independence	Clipping an action camera onto a drone — it rides along and records, it never touches the controls.

The close, if you get one line

"Every number we showed you was measured on this hardware — including the two bugs we found in our own system and the model we tested and chose not to switch to."

All figures traceable to docs/SaResQ_Model_Architecture_Report.pdf and docs/paper/saresq_paper.pdf · repository commit 7f1372a · 223 tests passing.